

Lecture-28
Mathematics 1 (15B11MA111)

CO [C105.5]

Module: Laplace Transforms

Topics to be covered

- ❖ Laplace Transform.
- ❖ Properties of Laplace Transforms.
- ❖ Laplace Transform of Derivatives
- ❖ Differentiation theorem or Multiplication by t^n
- ❖ Integration theorem or Division by t
- ❖ Laplace Transform of Integrals

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Definition

The **Laplace Transform** of a function $f(t)$ defined on $0 \leq t \leq \infty$ is denoted by $F(s)$ and is given by

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt = L\{f(t)\}$$

where s is a complex constant with real part greater than zero, and L is the **Laplace Transform operator**.

If s is real then $s > 0$.

Conditions for the Existence of a Laplace Transform of $f(t)$

1. $f(t)$ is piecewise continuous on $0 \leq t < \infty$.
2. $f(t)$ is of exponential order as $t \rightarrow \infty$.

That is there exist real constants K and T such that

$$|f(t)| \leq Ke^{at} \quad \text{for all } t > T$$

Note: *Conditions 1 and 2 are sufficient, but not necessary*

Example: Find the Laplace transform $F(s)$ of $f(t) = A t$.

Solution: The Laplace transform $F(s)$ of $f(t)$ is given by

$$F(s) = L\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

$$\therefore F(s) = \int_0^{\infty} A t e^{-st} dt = A \int_0^{\infty} t e^{-st} dt$$

Integrating by parts

$$\begin{aligned} F(s) &= A \left[t \frac{e^{-st}}{-s} \Big|_0^{\infty} - \int_0^{\infty} 1 \frac{e^{-st}}{-s} dt \right] = A(0 - 0) + \frac{A}{s} \int_0^{\infty} e^{-st} dt = \frac{A}{s} \frac{e^{-st}}{-s} \Big|_0^{\infty} \\ &= \frac{A}{s} \frac{e^{-st}}{-s} \Big|_0^{\infty} = \frac{A}{s^2} \end{aligned}$$

Example: Find the Laplace transform $F(s)$ of $f(t) = e^{at}$.

Solution: The Laplace transform $F(s)$ of $f(t)$ is given by

$$F(s) = L\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

$$\therefore F(s) = \int_0^{\infty} e^{at} e^{-st} dt = \int_0^{\infty} e^{-(s-a)t} dt$$

$$= \left| \frac{e^{-(s-a)t}}{-(s-a)} \right|_{t=0}^{t=\infty}$$

$$= \frac{0}{-(s-a)} - \frac{1}{-(s-a)}$$

(Only if $s > a$)

$$\therefore F(s) = L\{e^{at}\} = \frac{1}{s-a}$$

Laplace Transforms Table

$$f(t) \quad F(s) = L\{f(t)\}$$

$$1 \quad \frac{1}{s}$$

$$t \quad \frac{1}{s^2}$$

$$t^n \quad \frac{n!}{s^{n+1}}$$

$$e^{at} \quad \frac{1}{s-a}$$

$$te^{at} \quad \frac{1}{(s-a)^2}$$

$$f(t) \quad F(s) = L\{f(t)\}$$

$$\sin at \quad \frac{a}{s^2 + a^2}$$

$$\cos at \quad \frac{s}{s^2 + a^2}$$

$$\sinh at \quad \frac{a}{s^2 - a^2}$$

$$\cosh at \quad \frac{s}{s^2 - a^2}$$

Properties of Laplace Transforms

I. Linearity Property:

If a and b are constants and $f(t)$, $g(t)$ are two functions having Laplace Transforms $F(s)$, $G(s)$ respectively, then

$$L\{a f(t) \pm b g(t)\} = a L\{f(t)\} \pm b L\{g(t)\} = a F(s) \pm b G(s)$$

By definition, we have

$$\begin{aligned} L\{a f(t) + b g(t)\} &= \int_0^{\infty} [a f(t) + b g(t)] e^{-st} dt \\ &= a \int_0^{\infty} f(t) e^{-st} dt + b \int_0^{\infty} g(t) e^{-st} dt \\ &= a L\{f(t)\} + b L\{g(t)\} \\ &= a F(s) + b G(s) \end{aligned}$$

II. First Shifting Property:

If $L\{f(t)\} = F(s)$, then $L\{e^{at}f(t)\} = F(s-a)$, $a < s$ if real.

By definition, we have

$$L\{e^{at} f(t)\} = \int_0^{\infty} e^{at} f(t) e^{-st} dt = \int_0^{\infty} f(t) e^{-(s-a)t} dt = F(s-a)$$

III. Change of scale Property:

If $L\{f(t)\} = F(s)$, then $L\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$

By definition, we have

$$L\{f(at)\} = \int_0^{\infty} f(at) e^{-st} dt \quad \text{Put } at = u \text{ then } a dt = du$$

$$L\{f(at)\} = \int_0^{\infty} f(u) e^{-\frac{su}{a}} \frac{du}{a} = \frac{1}{a} \int_0^{\infty} f(u) e^{-\frac{su}{a}} du = \frac{1}{a} F\left(\frac{s}{a}\right)$$

Laplace Transform of Derivatives

If $L\{f(t)\} = F(s)$ and $f'(t)$ is continuous, show that

$$L\left\{\frac{d}{dt} f(t)\right\} = sF(s) - f(0)$$

$$L\left\{\frac{d^2}{dt^2} f(t)\right\} = s^2 F(s) - sf(0) - f'(0)$$

Proof: We have
$$\begin{aligned} L\left\{\frac{d}{dt} f(t)\right\} &= \int_0^{\infty} e^{-st} \frac{d}{dt} f(t) dt \\ &= e^{-st} f(t) \Big|_0^{\infty} - \int_0^{\infty} -se^{-st} f(t) dt \\ &= 0 - f(0) + sL\{f(t)\} = sL\{f(t)\} - f(0) \end{aligned}$$

$$\begin{aligned} L\left\{\frac{d^2}{dt^2} f(t)\right\} &= L\left\{\frac{d}{dt} \left(\frac{d}{dt} f(t)\right)\right\} = sL\left\{\frac{d}{dt} f(t)\right\} - f'(0) \\ &= s(sL\{f(t)\} - f(0)) - f'(0) = s^2 L\{f(t)\} - sf(0) - f'(0) \end{aligned}$$

For n^{th} derivative we have

$$\begin{aligned} L\left\{\frac{d^n}{dt^n} f(t)\right\} &= s^n L\{f(t)\} - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0) \\ &= s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0) \end{aligned}$$

Example: Find the Laplace transform $F(s)$ of $f(t) = \cos^2 \omega t$

Solution:

First Approach: From definition

$$\begin{aligned} F(s) &= \int_0^{\infty} f(t) e^{-st} dt = \int_0^{\infty} \cos^2 \omega t e^{-st} dt \\ &= \int_0^{\infty} \frac{1 + \cos 2\omega t}{2} e^{-st} d\tau = \frac{1}{2} \left(\frac{1}{s} + \frac{s}{s^2 + 4\omega^2} \right) = \frac{s^2 + 2\omega^2}{s(s^2 + 4\omega^2)} \end{aligned}$$

Second Approach: Using known results

We shall solve it by two ways.

$$(a) \quad F(s) = L\{f(t)\} = L\{\cos^2 \omega t\}$$

$$= L\left\{\frac{1}{2}(1 + \cos 2\omega t)\right\} = \frac{1}{2} \left(\frac{1}{s} + \frac{s}{s^2 + 4\omega^2} \right)$$

$$(b) \quad f(t) = \cos^2 \omega t \quad f(0) = 1$$

$$f'(t) = -2\omega \cos \omega t \sin \omega t = -\omega \sin 2\omega t$$

We have

$$L\{f'(t)\} = sL\{f(t)\} - f(0)$$

$$L\{-\omega \sin 2\omega t\} = sL\{f(t)\} - 1$$

$$-\omega \frac{2\omega}{s^2 + 4\omega^2} = sL\{f(t)\} - 1$$

$$sL\{f(t)\} = 1 - \omega \frac{2\omega}{s^2 + 4\omega^2} = \frac{s^2 + 4\omega^2 - 2\omega^2}{s^2 + 4\omega^2} = \frac{s^2 + 2\omega^2}{s^2 + 4\omega^2}$$

$$L\{\cos^2 \omega t\} = \frac{s^2 + 2\omega^2}{s(s^2 + 4\omega^2)}$$

Example: Find the Laplace transform $F(s)$ of $f(t) = \cosh t$

Solution: We have

$$\begin{aligned} f(t) = \cosh t &= \frac{e^t + e^{-t}}{2} \\ \therefore L\{f(t)\} &= L\left\{\frac{1}{2}(e^t + e^{-t})\right\} \\ &= \frac{1}{2}(L\{e^t\} + L\{e^{-t}\}) \\ &= \frac{1}{2}\left(\frac{1}{s-1} + \frac{1}{s+1}\right) \\ &= \frac{1}{2}\left(\frac{s+1+s-1}{s^2-1}\right) = \frac{s}{s^2-1} \end{aligned}$$

Example: Find the Laplace transform $F(s)$ of $f(t) = \sin 2t \cos t$

Solution: Applying the identity

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

we obtain

$$\begin{aligned} f(t) = \sin 2t \cos t &= \frac{1}{2} (\sin 3t + \sin t) \\ \therefore F(s) = L\{\sin 2t \cos t\} &= \frac{1}{2} (L\{\sin 3t\} + L\{\sin t\}) \\ &= \frac{1}{2} \left(\frac{3}{s^2 + 9} + \frac{1}{s^2 + 1} \right) \\ &= \frac{1}{2} \left(\frac{3(s^2 + 1) + s^2 + 9}{(s^2 + 9)(s^2 + 1)} \right) \\ &= \frac{2s^2 + 6}{(s^2 + 9)(s^2 + 1)} \end{aligned}$$

Example: Find the Laplace transform $F(s)$ of $f(t) = t \sin at$

Solution:

$$f(t) = t \sin at \qquad f(0) = 0$$

$$f'(t) = \sin at + at \cos at \qquad f'(0) = 0$$

$$f''(t) = 2a \cos at - a^2 t \sin at = 2a \cos at - a^2 f(t)$$

$$L\{f''(t)\} = L\{2a \cos at - a^2 f(t)\}$$

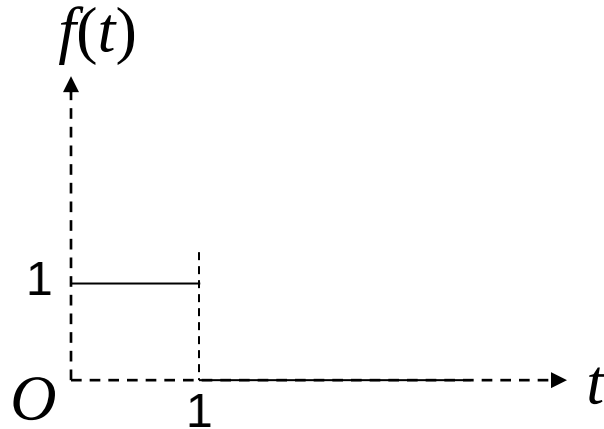
$$L\{f''(t)\} = 2aL\{\cos at\} - a^2L\{f(t)\}$$

$$s^2L\{f(t)\} - sf(0) - f'(0) = 2a \frac{s}{s^2 + a^2} - a^2L\{f(t)\}$$

$$(s^2 + a^2)L\{f(t)\} = \frac{2as}{s^2 + a^2}$$

$$\therefore F(s) = L\{f(t)\} = \frac{2as}{(s^2 + a^2)^2}$$

Example: Find the Laplace transform graph of



Solution: From the graph

$$f(t) = 1 \text{ for } 0 \leq t \leq 1$$

$$f(t) = 0 \text{ for } 1 < t$$

$$\begin{aligned} \therefore L\{f(t)\} &= \int_0^{\infty} f(t)e^{-st} dt \\ &= \int_0^1 (1)e^{-st} dt + \int_1^{\infty} (0)e^{-st} dt \end{aligned}$$

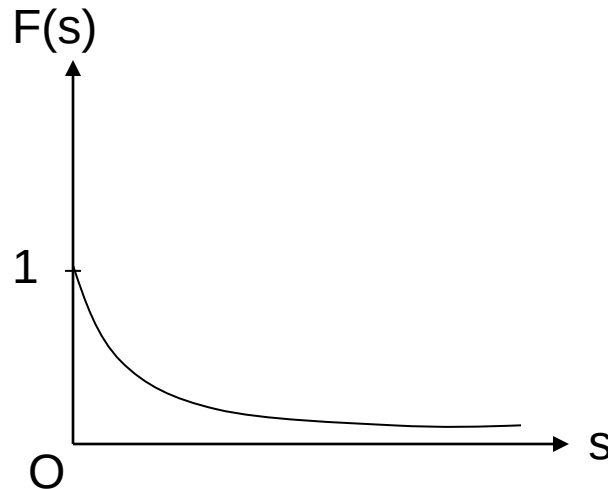
$$L\{f(t)\} = \int_0^1 e^{-st} dt = \left| \frac{e^{-st}}{-s} \right|_{t=0}^{t=1} = \frac{e^{-s}}{-s} - \frac{1}{-s} = \frac{1 - e^{-s}}{s}$$

To draw the graph of

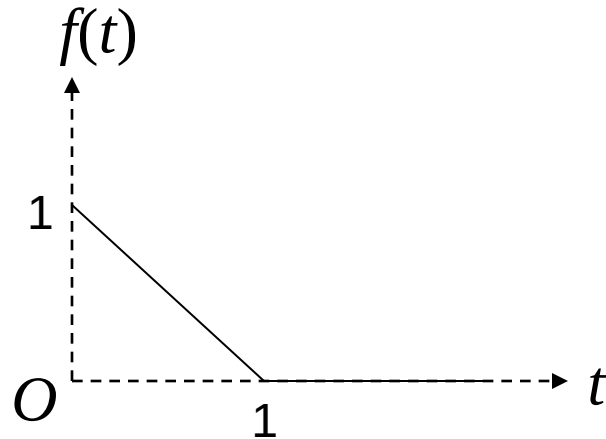
$$F(s) = L\{f(t)\} = \frac{1 - e^{-s}}{s}$$

As $s \rightarrow 0$, $F(s) \rightarrow 1$ As $s \rightarrow \infty$, $F(s) \rightarrow 0$

For $0 \leq s < \infty$, $1 \geq F(s) \geq 0$



Example: Find the Laplace transform graph of



Solution: The graph of $f(t)$ is the a straight line through the points $(0, 1)$ and $(1, 0)$ and t axis for $t > 0$.

Therefore the function $f(t)$ can be written as

$$\text{For } 0 \leq t \leq 1 \quad f(t) = 1 - t$$

$$\text{and for } t \geq 1 \quad f(t) = 0 \quad \text{or} \quad f(t) = \begin{cases} 1-t & \text{for } 0 \leq t \leq 1 \\ 0 & \text{for } t \geq 1. \end{cases}$$

$$L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt = \int_0^1 (1-t)e^{-st} dt + \int_1^{\infty} (0)e^{-st} dt$$

$$\therefore L\{f(t)\} = \int_0^1 (1-t)e^{-st} dt$$

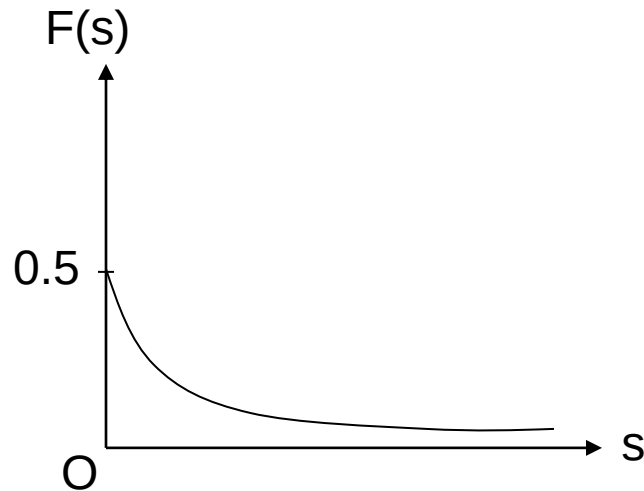
Integrating by parts, we get

$$\begin{aligned} L\{f(t)\} &= \left| (1-t) \frac{e^{-st}}{-s} \right|_{t=0}^{t=1} - \int_0^1 (0-1) \frac{e^{-st}}{-s} dt \\ &= \left(0 - \frac{1}{-s} \right) - \frac{1}{s} \int_0^1 e^{-st} dt = \frac{1}{s} - \frac{1}{s} \left| \frac{e^{-st}}{-s} \right|_{t=0}^{t=1} \\ &= \frac{1}{s} + \frac{1}{s^2} (e^{-s} - 1) = \frac{1}{s^2} (e^{-s} + s - 1) \\ F(s) &= \frac{1}{s^2} (e^{-s} + s - 1) \quad 0 \leq s < \infty \end{aligned}$$

To draw the graph of

$$F(s) = \frac{1}{s^2} (e^{-s} + s - 1) \quad 0 \leq s < \infty$$

As $s \rightarrow 0$, $F(s) \rightarrow 0.5$ As $s \rightarrow \infty$, $F(s) \rightarrow 0$



Differentiation theorem or Multiplication by t^n

$$L\{t^n f(t)\} = (-1)^n F^{(n)}(s)$$

Proof: We know that

$$F(s) = L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

Differentiating both sides with respect to s

$$\begin{aligned} \frac{d}{ds} F(s) &= \frac{d}{ds} \int_0^{\infty} f(t)e^{-st} dt = \int_0^{\infty} f(t) \frac{d}{ds} e^{-st} dt \\ &= - \int_0^{\infty} t f(t) e^{-st} dt = -L\{t f(t)\} \end{aligned}$$

$$\therefore L\{t f(t)\} = -\frac{d}{ds} F(s)$$

Thus the result holds for $n = 1$.

$$\begin{aligned}
 L\{t^2 f(t)\} &= L\{t[tf(t)]\} \\
 &= -\frac{d}{ds}[L\{tf(t)\}] = -\frac{d}{ds}\left[-\frac{d}{ds}L\{f(t)\}\right] = \frac{d^2}{ds^2}F(s)
 \end{aligned}$$

Thus the result holds for $n=2$ also.

Now assume that the result holds for $n = m$ i.e.

$$L\{t^m f(t)\} = (-1)^m F^{(m)}(s)$$

$$L\{t^{m+1} f(t)\} = L\{t^m [tf(t)]\}$$

$$= (-1)^m \frac{d^m}{ds^m} [L\{tf(t)\}] \quad \text{As the result holds for } n = m.$$

$$= (-1)^m \frac{d^m}{ds^m} \left[-\frac{d}{ds} F(s) \right] = (-1)^{m+1} F^{(m+1)}(s)$$

Hence the result holds for all $n= 1, 2, 3, \dots\dots\dots$

Integration theorem or Division by t

$$L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(s) ds$$

Proof: We know that

$$F(s) = L\{f(t)\} = \int_0^\infty f(t)e^{-st} dt$$

Integrating both sides with respect to s from s to ∞ .

$$\begin{aligned} \int_s^\infty F(s) ds &= \int_s^\infty \left[\int_0^\infty f(t)e^{-st} dt \right] ds = \int_0^\infty \left[\int_s^\infty f(t)e^{-st} ds \right] dt \\ &= \int_0^\infty f(t) \left[\int_s^\infty e^{-st} ds \right] dt = \int_0^\infty f(t) \left[\frac{e^{-st}}{-t} \right]_s^\infty dt \\ &= \int_0^\infty f(t) \left[\frac{0}{-t} - \frac{e^{-st}}{-t} \right] dt = \int_0^\infty \frac{f(t)}{t} e^{-st} dt = L\left\{\frac{f(t)}{t}\right\} \end{aligned}$$

Laplace Transform of Integrals:

If $L\{f(t)\} = F(s)$, show that $L\left\{\int_0^t f(u) du\right\} = \frac{F(s)}{s}$

$$\text{Let } \frac{dg(t)}{dt} = f(t) \quad \text{Then } \int_0^t f(u) du = g(t) - g(0)$$

$$\begin{aligned} L\left\{\int_0^t f(u) du\right\} &= L\{g(t) - g(0)\} \\ &= L\{g(t)\} - L\{g(0)\} = L\{g(t)\} - \frac{g(0)}{s} \end{aligned} \quad (1)$$

$$\begin{aligned} L\left\{\frac{dg(t)}{dt}\right\} &= L\{f(t)\} & sL\{g(t)\} - g(0) &= F(s) \\ \therefore L\{g(t)\} - \frac{g(0)}{s} &= \frac{F(s)}{s} \end{aligned} \quad (2)$$

Combining (1) and (2) we get $L\left\{\int_0^t f(u) du\right\} = \frac{F(s)}{s}$

Practice Questions:

1. Find the Laplace Transform of $\{te^{-t}\sin 3t\}$

Ans. $\frac{6(s+1)}{(s^2+2s+10)^2}$

2. Find the Laplace Transform of $\left\{\frac{1-e^{-t}}{t}\right\}$

Ans. $\log \frac{s}{s-1}$

3. Evaluate $L\left\{\int_0^t \sin t dt\right\}$

Ans. $\frac{1}{s(s^2+1)}$

Thank you